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Enhancing nitrogen status evaluation using qualitative indicators: a case study in Matsushima Bay, Japan

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Abstract

Identifying nitrogen dynamics in coastal ecosystems remains a key challenge for environmental monitoring and management. In this study, nitrogen distribution patterns were examined in Matsushima Bay, a semi-enclosed coastal system, and its adjacent areas. The results revealed the utility of the natural abundance of ^{15}N in all dissolved nitrogen ($\delta^{15}N$-TDN) as a qualitative indicator of coastal nutrient sources and status. Quantitative analyses showed substantial nitrogen loading from the Sunaoshi and Naruse rivers, but limited downstream detectability, likely because of rapid attenuation and mixing. Notably, $\delta^{15}N$-TDN measurements indicated persistent isotopic differences between the mouth of Matsushima Bay and Sendai Bay, underscoring their enduring disparities despite their geographic proximity. Furthermore, $SUVA_{254}$ (specific UV absorbance at 254 nm), a parameter derived from dissolved organic carbon (DOC), outperformed bulk DOC concentration data by more effectively capturing terrestrial organic matter influences. These findings highlight the value of $\delta^{15}N$-TDN and $SUVA_{254}$ as refined qualitative indicators for delineating nuanced variations in nutrient sources and status, particularly in areas where treated wastewater contributes substantially. They reveal that terrestrial nitrogen inputs, while substantial near river mouths, attenuate rapidly within the bay, leaving detectable isotopic and compositional signals in downstream waters. Furthermore, the findings emphasize the value of integrated qualitative and quantitative approaches for better nutrient status assessment and for more effective environmental management strategies.

Keywords $\delta^{15}N$-TDN, $SUVA_{254}$, Coastal ecosystems, Matsushima Bay, Riverine influence

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1 Introduction

Coastal ecosystems are increasingly threatened by multiple anthropogenic stressors (Halpern et al. 2015; Solé Figueras et al. 2024). Among these, the excessive input of nutrients such as nitrogen (N) is particularly destructive, leading to widespread degradation by processes including eutrophication, hypoxia, and disruptions to ecosystem functioning (Rabalais 2002; Howarth 2008; Breitburg et al. 2018). Therefore, elucidating nutrient inputs from land to coastal ecosystems, as well as their effects, is necessary for the conservation and management of these systems (Herbert 1999).

Various measurements are employed frequently to assess the N status of coastal ecosystems, using both

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water samples and biological samples. For water samples, these include concentrations of dissolved inorganic N (DIN) such as ammonium (NH_4^+) (Dugdale and Goering 1967; Eppley et al. 1979), nitrate (NO_3^-) (Dortch 1990; Rabalais 2002), dissolved organic nitrogen (DON) (Bronk et al. 1994, 2007; Berman and Bronk 2003), total dissolved nitrogen (TDN; DIN + DON), and particulate organic nitrogen (PON) (Bronk and Ward 2005), as well as total nitrogen (TN; TDN + PON). For biological samples, common measurements include the N contents or carbon-to-nitrogen (C/N) ratios of algae, seagrasses, and bivalves (Fukumori et al. 2008). These parameters are typically examined in conjunction with measurements of water temperature, salinity, and dissolved oxygen (DO). Among these, DON and TDN are increasingly recognized as ecologically important nitrogen pools that support both phytoplankton and heterotrophic microbial activity in coastal ecosystems (Seitzinger et al. 2002; Zhang et al. 2022). Numerous studies have demonstrated that bioavailable DON from terrestrial sources can substantially contribute to coastal eutrophication and even facilitate the development of harmful algal blooms (Seitzinger and Sanders 1997; Li et al. 2020; Zhang et al. 2022). These findings underscore the importance of tracing the origin and transformation of TDN components in coastal environments.

These various concentration-based indicators have been widely applied in coastal studies worldwide, primarily because they are methodologically straightforward. However, the high biological demand for N often causes rapid (hours to days) and wide fluctuations in measured N concentrations in coastal ecosystems. When longer-term (months to years) assessments must be conducted, macroalgal N content is commonly used. However, macroalgae are typically limited in spatial distribution within the bay unless artificially deployed, which limits their utility in assessing nutrient status at the bay-wide scale. Moreover, their N content is also influenced by biological processes such as photosynthesis, N uptake physiology, and species-specific traits. As a result, concentration-based metrics alone are insufficient to fully assess the N status of coastal systems, particularly with respect to land-derived N (Beusen et al. 2016).

One established approach is the measurement of the natural abundance of ^{15}N ($\delta^{15}N$). Although $\delta^{15}N$ measurements are more labor-intensive than concentration measurements, $\delta^{15}N$ provides integrative information related to the production and consumption processes of N, as well as its sources (Robinson 2001; Fry et al. 2003). Because terrestrial sources of N often have higher $\delta^{15}N$ than oceanic N sources, $\delta^{15}N$ measurements can be extremely effective for tracing the fate of land-derived N (Reichwaldt and Ghadouani 2016; Samper-Villarreal et al. 2020). $\delta^{15}N$ values in primary producers (e.g., PON, macroalgae, seagrasses) and herbivores (e.g., bivalves) have been widely used to understand N utilization and trophic relationships in coastal ecosystems (Samper-Villarreal et al. 2020). However, such indicators often remain insensitive to short-term or small-scale N inputs, such as pulsed or continuous low-level terrestrial N loads, indicating that more sensitive parameters are needed.

To attain higher sensitivity in tracing land-derived N, determining the $\delta^{15}N$ of dissolved N compounds is crucially important. Nevertheless, DIN concentrations in coastal waters are occasionally low for $\delta^{15}N$ measurements. Although DIN is invaluable, obtaining adequate N for $\delta^{15}N$ measurements has posed difficult challenges. In the case of NH_4^+ in coastal water, which is highly bioavailable and which is considered particularly sensitive, early methods relied on Kjeldahl distillation necessitating accumulation of high-concentration samples (Velinsky et al. 1989). An improved diffusion method (Holmes et al. 1998) combined with elemental analyzer–isotope ratio mass spectrometry (EA-IRMS) has enabled analysis of low concentrations, but documented applications remain limited (McClelland and Valiela 1998; Middelburg and Nieuwenhuize 2001; Fry et al. 2003; Dillon et al. 2007; Dillon and Chanton 2008; Sugimoto et al. 2009). For NO_3^-, development of analytical methods with N_2O as a final analyte (Sigman et al. 2001; Casciotti et al. 2002; McIlvin and Altabet 2005) has reduced the required N amount dramatically, from a few micromoles (with diffusion–EA-IRMS) to as low as approx. 20 nmol-N (approx. 0.28 μg-N), thereby supporting important progress (e.g., Hosono et al. 2013; Archana et al. 2018; Johannsen et al. 2008). Although analogous techniques for converting NH_4^+ to N_2O have also been developed (Houlton et al. 2006; Zhang et al. 2007), they require concentrations of several micromolar N, thereby limiting their utility in many marine environments.

Despite being less bioavailable than DIN, DON is nonetheless expected to serve as an indicator of land-derived N. However, $\delta^{15}N$ analysis of DON remains extremely challenging for coastal samples where preconcentration is not straightforward. There has been little subsequent progress since the application of dialysis by Feuerstein et al. (1997). A method to measure $\delta^{15}N$ of TDN ($\delta^{15}N$-TDN) via wet oxidation (Knapp et al. 2005) has expanded the possibilities for determining $\delta^{15}N$-TDN in coastal waters. Nonetheless, deriving DON $\delta^{15}N$ values requires subtracting DIN from TDN, which reduces measurement precision and which necessitates considerable effort to quantify both DIN concentrations and $\delta^{15}N$. Although direct $\delta^{15}N$ measurements of solid-phase extraction DON have been introduced recently (Broek

et al. 2023), this approach is not yet suitable for large-scale monitoring.

Considering the great number of sample measurements needed for monitoring and regarding the necessity for a distinct $\delta^{15}N$ signal to trace terrestrial N, this study proposes using TDN $\delta^{15}N$. Although TDN includes various N forms and is not highly specific, its $\delta^{15}N$ value may still retain the isotopic signature of terrestrial N, especially in samples where DON dominates the TDN pool. This feature is particularly advantageous in coastal waters where NO_3^- and NH_4^+ concentrations are too low for direct $\delta^{15}N$ measurements. Therefore, acknowledging that $\delta^{15}N$-TDN reflects both DIN and DON, we investigate whether the high $\delta^{15}N$ characteristics of land-derived N are apparent even in areas with insufficient DIN for isotopic analysis, thereby enabling the tracing of land-derived N. $\delta^{15}N$-TDN has already been applied across diverse environments, including ocean systems (Granger et al. 2011; Alkhatib et al. 2012; Thibodeau et al. 2013) terrestrial ecosystems (Mnich and Houlton 2016; Savard et al. 2021; Alvarez et al. 2024), the atmosphere (Liu et al. 2017), and freshwater environments (Jeong et al. 2022; Chaves et al. 2021). In these studies, $\delta^{15}N$ of TDN was used either to estimate the $\delta^{15}N$ of DON (e.g., Liu et al. 2017), or, under low DIN conditions, to represent the isotopic composition of reduced dissolved N (NH_4^+ + DON) (Granger et al. 2011; Alkhatib et al. 2012) or DON alone (Thibodeau et al. 2013; Chaves et al. 2021).

This study introduces $\delta^{15}N$-TDN as a potential solution to this challenge for the better application of $\delta^{15}N$. By analyzing $\delta^{15}N$-TDN, limitations posed by low concentrations of DIN can be mitigated because of the higher concentration of TDN (DIN + DON) providing a sufficient concentration for $\delta^{15}N$ analysis. Dissolved organic N can account for 20% to 90% of the total N loading to coastal areas, further aiding in reaching the detection limit for TDN analysis (Seitzinger and Sanders 1997). Traditionally, $\delta^{15}N$ analysis of TDN or DON in freshwater and seawater has been challenging because of its low concentration and high salt accumulation during pretreatment for analyses (Feuerstein et al. 1997), but recent developments of "oxidation–denitrifier" (O-D) method (Knapp et al. 2005; Moretti et al. 2024) have opened a new window to measure $\delta^{15}N$-TDN with much lower concentrations.

Additionally, specific UV absorbance at 254 nm ($SUVA_{254}$), widely used as a surrogate measure for dissolved organic carbon (DOC) aromaticity to infer the composition and sources of dissolved organic matter (Weishaar et al. 2003), has been applied in various aquatic environments including rivers (Spencer et al. 2012), lakes (Selberg et al. 2011), and coastal systems (Wagner et al. 2015), was applied in this work. The present study, using Matsushima Bay as a case, was conducted to demonstrate the benefits of employing $\delta^{15}N$-TDN and $SUVA_{254}$ as qualitative indicators in the field of coastal nutrient dynamics. These indicators have the potential to provide more precise and comprehensive information than conventional nutrient concentration analyses can. Furthermore, the integration of qualitative indicators with traditional nutrient analyses might offer a more complete picture of nutrient sources, transformations, and overall ecosystem health.

2 Materials and methods

2.1 Site description

Matsushima Bay is an enclosed bay of approximately 35.3 km^2 area, with an average depth of about 4 m and observed depths ranging from 0.1 to 12 m based on long-term monitoring data, and a bay mouth width of approximately 1.7 km (International EMECS Center 2001). It is located in the central part of Sendai Bay, i.e., a shallow semicircular bay with maximum depth of approx. 50 m, situated in the northeastern part of Honshu Island, Japan (Figs. 1 and S1a, b), and was minimally influenced by offshore currents during our September survey, as indicated by the temperature–salinity diagram analysis (Fig. S2). Matsushima Bay receives direct inflow from one river (the Takagi River) and additional inflow via narrow canals from two other rivers that flow into Sendai Bay (Fig. 1). Specifically, the Sunaoshi River influences Matsushima Bay mainly through the Sunaoshi-Teizan Canal, which also receives effluent from a large-scale sewage treatment plant (Sen-en Wastewater plant; Fig. 1), while the Naruse River contributes both through the Touna Canal and its main outlet, which is only partially separated from the bay by Miyato Island. These hydrological features indicate that the typical interpretation of material cycles in an inner bay, based solely on salinity, cannot be directly applied to Matsushima Bay. Although Matsushima Bay is famous for the cultivation of oyster and seaweed (Okumura et al. 2023), its N status is not nutrient-rich with low DIN concentrations of 0.1–20.4 μmol L^{-1} for surface water and 0.2–5.9 μmol L^{-1} for bottom water during 2009–2015, with large spatial variation (Okumura et al. 2021) where separate $\delta^{15}N$ analyses for NH_4^+ and NO_3^- are difficult to conduct.

2.2 Water sampling in Matsushima Bay and Sendai Bay

We collected seawater samples from Matsushima Bay and Sendai Bay for nutrient and stable isotopic analysis from September 30 through October 2, 2023, as a part of the regular monitoring campaign conducted by Miyagi Prefecture Fisheries Technology Institute (MPFTI). On September 30 (cloudy weather), water samples in Sendai Bay were collected from 9 stations (SD1 – SD9;

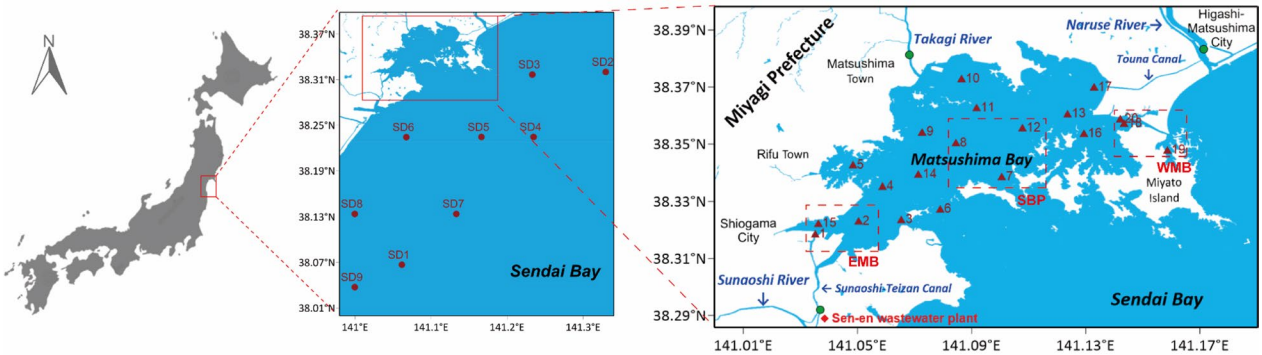

Fig. 1 Map of sampling sites. In all, 20 stations were established in Matsushima Bay (MS) and 9 stations in Sendai Bay (SD). Green circles of the right panel denote the sampling sites of three rivers. For simplicity, station labels in Matsushima Bay are shown as △1–20, corresponding to MS1–MS20. MS1, MS2, and MS15 were categorized as being in Western Matsushima Bay (WMB, near the Sunaoshi-Teizan Canal), whereas MS18, MS19, and MS20 were categorized as in Eastern Matsushima Bay (EMB, influenced by the Naruse River). Stations MS7, MS8, and MS12 were classified as being in proximity to Sendai Bay (SBP). The location of the Sen-en wastewater treatment plant is also shown as a red diamond. These three subregions are outlined with red dashed boxes. The map was compiled using the Tiles of Geospatial Information Authority of Japan (https://maps.gsi.go.jp/)

Fig. 1), using Niskin bottles, with 3–5 depths sampled per station depending on the maximum depth (generally 20–30 m; samples were taken every ~ 10 m and additionally at the bottom when the remaining depth was < 10 m; see Table S3 for details). On October 2 (sunny weather), water samples from both surface and bottom layers were collected from 20 stations in Matsushima Bay (MS1 – MS20; Fig. 1). For certain sampling stations in Matsushima Bay, which were located near specific features, we categorized them to facilitate detailed analysis. Stations MS1, MS2, and MS15 were categorized as being in Western Matsushima Bay (WMB; Fig. 1), near the Sunaoshi-Teizan Canal and possibly affected by high N loading from a wastewater plant (Okumura et al. 2021). The Naruse River contributes substantial nitrogen input to Matsushima Bay because of its higher discharge rates (Table S1) and river water, which, respectively, accounted for 8.2%, 16.9%, and 9.6% of MS18, MS19, and MS20 based on salinity (Supplementary text). Therefore, we categorized MS18, MS19, and MS20 as being in Eastern Matsushima Bay (EMB; Fig. 1), influenced by the Naruse River with low salinity (Fig. S1c and d). Stations MS7, MS8, and MS12 were classified as being in proximity to Sendai Bay (SBP; Fig. 1), to highlight their closer connection with Sendai Bay (Okumura et al. 2021). Although MS6 is located near Sendai Bay, its nutrient concentrations were significantly higher than those of the SBP stations (Table S2), likely because of a strong influence from the Sunaoshi-Teizan Canal. Therefore, it was not included in the SBP category.

Temperature, salinity, and DO concentrations were measured vertically from the surface to the bottom using a conductivity–temperature–depth (CTD) profiler (RINKO Profiler; JFE Advantech Co. Ltd., Nishinomiya, Japan) by MPFTI. Water samples collected using a VAN DORN water sampler were filtered using a glass fiber filter (GF/C) in the laboratory as a part of the regular nutrient monitoring conducted every two months, whereas samples for δ^{15}N-TDN, TDN, and DOC were filtered on the ship with glass fiber filter (GF/F; 450 °C precombusted, 4 h). Then the filtrate was chilled, frozen at −20 °C, and shipped to Center for Ecological Research (CER), Kyoto University.

2.3 Water sampling for rivers running into Matsushima Bay

River samples were collected on September 28 and 29, 2023, from the Sunaoshi-Teizan Canal, Takagi, and Naruse rivers at each river mouth (Fig. 1). After water was collected using a bucket, it was filtered through GF/F filters. Then the filtrates were chilled and frozen before being shipped to CER.

2.4 Measurements of nutrient concentrations and $SUVA_{254}$

Concentration analysis of DIN for Matsushima Bay and Sendai Bay samples was conducted at MPFTI using a colorimetric method with an autoanalyzer (QuAAtro2-HR; BLTEC, Osaka, Japan). The obtained monitoring data can be referred from the MPFTI website: Miyagi Prefecture (2023a) for Matsushima Bay and Miyagi Prefecture (2023b) for Sendai Bay.

River water samples were analyzed for DIN and PO_4^{3-}-P concentrations using a flow injection analyzer (OG-FI-300S for $NO_3^- + NO_2^-$, OG-FI-3300NH for NH_4^+, OG-FI-3300PO for PO_4^{3-}-P; Ogawa & Co., Kobe, Japan) at Shiga Prefectural University. Both river and

seawater samples were analyzed for DOC and TDN using a total organic carbon (TOC) analyzer equipped with a Total Nitrogen Module (TOC-L; Shimadzu Corp., Kyoto, Japan) at the Field Science Education and Research Center (FSERC, Kyoto University).

$SUVA_{254}$ was measured using a spectrometer (UV 1240; Shimadzu Corp., Kyoto, Japan), and baseline correction was performed with ultrapure Milli-Q water. $SUVA_{254}$ was calculated by dividing the absorbance at $\lambda = 254$ nm (UV_{254}) measured with 1 cm quartz cell path length by the DOC concentration (mg L^{-1}), following the method described by Weishaar et al. (2003). The calculation is the following:

$$SUVA_{254}\left[\mathrm{L\ mg^{-1}Cm^{-1}}\right] = \frac{UV_{254}\left(\mathrm{cm^{-1}}\right)}{DOC\left(\mathrm{mg\ L^{-1}}\right)} \times 100\left(\frac{\mathrm{cm}}{\mathrm{m}}\right)$$

2.5 Isotopic measurements with O-D method

To digest TDN to NO_3^-, 4 mL of the filtrate was wet-oxidized with 1 mL of persulfate reagent (prepared by dissolving 1.52 g sodium hydroxide, 3 g boric acid, and 5 g potassium persulfate in 100 mL deionized water) in a capped test tube (Koba et al. 2010, 2012). In cases where the original TDN concentration in the water samples was high (e.g., river water), the filtrate was diluted before analysis. Conversion of the prepared NO_3^- to N_2O by denitrifying bacteria (Sigman et al. 2001) was conducted at CER (Kobayashi et al. 2021). The produced N_2O was introduced into stable isotope mass spectrometry coupled with a purge-and-trap preconcentration unit and a gas chromatograph (Kobayashi et al. 2021) at CER and Japan Agency for Marine–Earth Science and Technology (JAMSTEC). $\delta^{15}N$ is expressed in accordance with the international standard scale using the following equation:

$$\delta^{15}\mathrm{N} = 1000 \times \left[\left(R_{\mathrm{sample}}/R_{\mathrm{standard}}\right) - 1\right],$$

where R_{sample} is the $^{15}N/^{14}N$ ratio of the sample, and R_{standard} is that of atmospheric N. The raw $\delta^{15}N$ values were calibrated against laboratory standards: alanine ($\delta^{15}N = 22.7$‰; CERKU-02) and threonine ($\delta^{15}N = -2.9$‰; CERKU-05) (Tayasu et al. 2011). The precision and accuracy were confirmed via repeated analysis of glycine ($\delta^{15}N = 2.2$‰; CERKU-03), measured as 1.7 ± 0.5‰ (mean ± SD, $n = 4$). A comparison of TDN concentrations obtained using this method and those obtained using a TOC analyzer at FSERC showed close agreement, indicating the reliability and accuracy of both methods (Fig. S3).

To evaluate the effect of persulfate reagent on TDN oxidation, we tested a seawater sample with high TDN concentration under varying reagent levels from 0 to 10%. Both the TDN concentration and $\delta^{15}N$ values stabilized when the reagent proportion exceeded 5% (Fig. S4). Additionally, blank tests conducted with three replicates of 4 mL deionized water and 1 mL persulfate reagent showed a peak area that was only 3.3 ± 3.1% of the average peak area of seawater samples, confirming that the influence of the blank is minimal after blank correction.

2.6 Statistical analyses

Statistical analyses were conducted to compare qualitative and quantitative measurements between groups. Pairwise comparisons were performed using unpaired *t*-tests (for TDN, NH_4^+ concentrations, and $\delta^{15}N$-TDN values between the SBP region and Sendai Bay) and Mann–Whitney tests (for NO_3^- concentrations between the SBP region and Sendai Bay), depending on the normality of the data. For multiple comparisons across rivers, Matsushima Bay, and Sendai Bay, one-way ANOVA (for DOC concentrations and $\delta^{15}N$-TDN values) and Kruskal–Wallis tests (for TDN, NH_4^+, NO_3^- concentrations and $SUVA_{254}$ values) were applied, followed by Tukey's and Dunn's multiple comparisons tests, respectively, considering assumptions of variance homogeneity and data normality. Significance was inferred for $p < 0.05$. All analyses were performed using GraphPad Prism (GraphPad, San Diego, CA, USA).

3 Results

3.1 Nitrogen distribution in Matsushima Bay

Concentrations of TDN were found to be 57.4–548.5 μmol L^{-1} in rivers, 8.4–43.1 μmol L^{-1} in Matsushima Bay, and 4.9–19.0 μmol L^{-1} in Sendai Bay (Fig. 2a; Tables S2–S4), where the Sunaoshi-Teizan Canal had distinctively high TDN concentrations, with measured values of 327.1 and 548.5 μmol L^{-1}, compared to 57.4 and 72.3 μmol L^{-1} in the Takagi River, and 59.3 and 73.9 μmol L^{-1} in the Naruse River (Fig. 2c and Table S4). Concentrations of TDN in all three rivers were higher than those in Matsushima Bay ($p < 0.01$; Fig. 2a). Moreover, TDN concentrations in Matsushima Bay were higher than those in Sendai Bay ($p < 0.05$, Fig. 2a and S4a). The concentrations of TDN did not differ between surface and bottom waters in Matsushima Bay ($p = 0.48$; Table S2; Fig. S5b–c). Consequently, the WMB region of Matsushima Bay exhibited the highest TDN concentrations in both surface and bottom waters (Fig. 3a and b). Additionally, the EMB region represented another TDN hotspot (9.8, 19.8, and 17.1 μmol L^{-1}, respectively), but with lower TDN levels compared to those of the EMB region (Fig. 3a–b and S4b–S4c). In contrast, at the SBP region, TDN concentrations were not significantly different from those in Sendai Bay ($p = 0.55$, Fig. 4a).

Concentrations of NO_3^- were 27.9–366.8 μmol L^{-1} in rivers, 0.1–89.5 μmol L^{-1} in Matsushima Bay, and

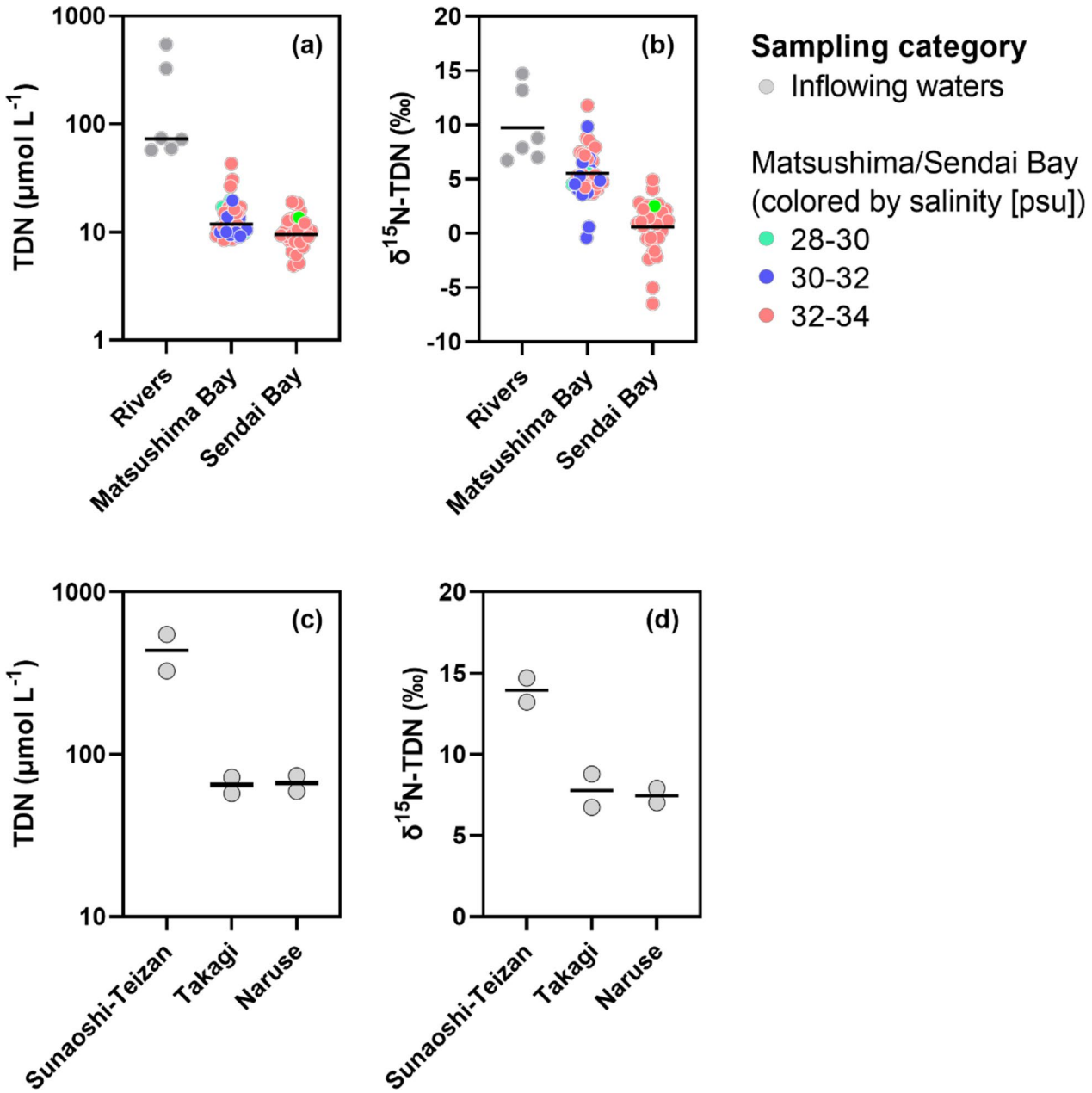

Fig. 2 Comparison of TDN concentrations and δ^{15}N-TDN values found for rivers, Matsushima Bay, and Sendai Bay: **a** TDN concentrations across the rivers and bays; **b** δ^{15}N-TDN values across the rivers and bays; **c** TDN concentrations specifically for the Sunaoshi-Teizan Canal, Takagi, and Naruse rivers; and **d** δ^{15}N-TDN values specifically for the Sunaoshi-Teizan Canal, Takagi, and Naruse rivers. The y-axis, representing TDN concentrations, is set on a logarithmic scale. Red lines represent the medians of the TDN concentrations in panels **a** and **c**, and the means of the δ^{15}N-TDN values in panels **b** and **d**. Gray circles in **a** and **b** indicate inflowing waters (rivers and canals), while colored circles represent Matsushima Bay and Sendai Bay stations categorized by salinity ranges (28–30, 30–32, 32–34 psu)

0–0.9 μmol L^{-1} in Sendai Bay (Fig. S6a; Tables S2–S4). Notably, the Sunaoshi-Teizan Canal exhibited distinctively high NO_3^- concentrations, with observed values of 290.0 and 368.1 μmol L^{-1}, compared to 27.9 and 48.7 μmol L^{-1} in the Takagi River and 50.7 and 59.3 μmol L^{-1} in the Naruse River (Fig. S6c and Table S4). Levels of NO_3^- in all three rivers were higher than those in Matsushima Bay ($p < 0.01$; Fig. S6a); NO_3^- concentrations in Matsushima Bay were higher than those in Sendai Bay ($p < 0.01$, Fig. S6a). Concentrations of NO_3^- did not differ between surface and bottom waters in Matsushima Bay ($p = 0.35$; Table S2 and Fig. S7a).

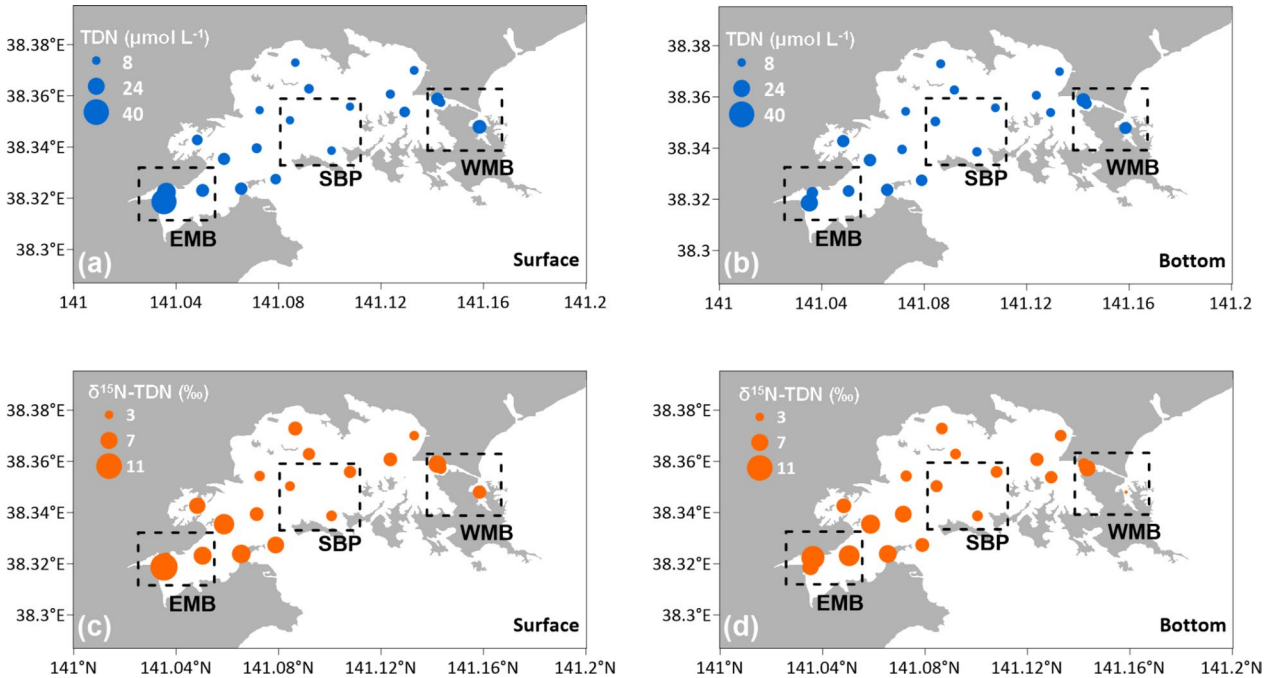

Fig. 3 Spatial distributions of TDN concentrations and δ^{15}N-TDN in Matsushima Bay: **a** surface TDN concentrations, **b** bottom TDN concentrations, **c** surface δ^{15}N-TDN values, and **d** bottom δ15N-TDN values. Circles at sampling stations represent the magnitudes of the values, with larger circles representing higher concentrations or larger values

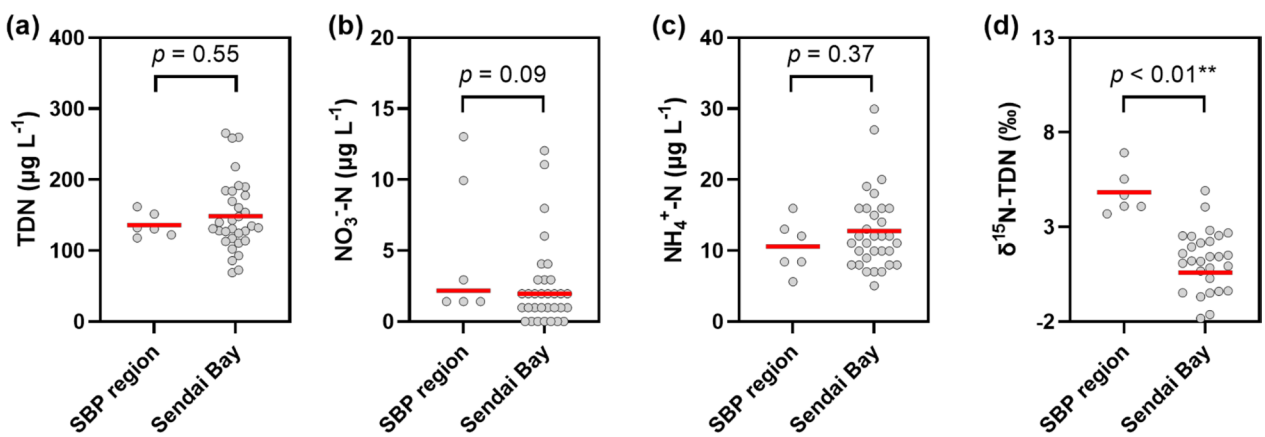

Fig. 4 Concentrations of various N species and δ^{15}N-TDN values found for the SBP region of Matsushima Bay (MS7, MS8, and MS12) and Sendai Bay: **a** TDN, **b** NO3⁻-N, **c** NH4⁺-N, and **d** δ^{15}N-TDN. The red lines represent the means of the TDN **a**, NH4⁺-N **c** concentrations, and δ^{15}N-TDN values **d**, and the median of the NO3⁻-N **b** concentrations

Concentrations of NH_4^+ were 2.8–176.8 μmol L^{-1} in rivers, 0.4–53.7 μmol L^{-1} in Matsushima Bay, and 0.4–2.1 μmol L^{-1} in Sendai Bay (Fig. S6b; Table S2-4). The Sunaoshi-Teizan Canal exhibited particularly high NH_4^+ concentrations, with measured values of 44.7 and 176.8 μmol L^{-1}, compared to 11.3 and 11.7 μmol L^{-1} in the Takagi River and 2.8 and 3.2 μmol L^{-1} in the Naruse River (Fig. S6d and Table S4). Levels of NH_4^+ in all three rivers were higher than those in Matsushima Bay ($p < 0.01$; Fig. S6b); NH_4^+ concentrations in Matsushima Bay were higher than those in Sendai Bay ($p < 0.01$, Fig. S6b). Concentrations of NH_4^+ did not differ between surface and bottom waters in Matsushima Bay ($p = 0.39$; Table S2 and Fig. S7b). The spatial distributions of NH_4^+ and NO_3^- in Matsushima Bay paralleled the distribution of TDN ($p < 0.01$; Fig. S8). As in the case for TDN, the

NH_4^+ and NO_3^- concentrations of the SBP region were found to have no significant difference from those in Sendai Bay ($p > 0.05$, Fig. 4b and c) reflecting a considerable degree of mixing of seawater between Matsushima Bay and Sendai Bay, making it difficult to detect river-derived effects based solely on N concentrations.

3.2 δ^{15}N-TDN distribution in Matsushima Bay

Results indicate δ^{15}N-TDN as 6.8–14.7‰ in rivers, −0.4–11.8‰ in Matsushima Bay, and −6.5 to 4.9 ‰ in Sendai Bay (Fig. 2b and Table S2-4), where the Sunaoshi-Teizan Canal had distinctively high δ^{15}N-TDN concentrations (13.2–14.7‰ for Sunaoshi-Teizan Canal, 6.8–8.8‰ for Takagi River, and 7.0–7.9% for Naruse River; Fig. 2d and Table S4). The δ^{15}N-TDN levels in all three rivers were higher than those in Matsushima Bay ($p < 0.01$; Fig. 2b); δ^{15}N-TDN values in Matsushima Bay were higher than those in Sendai Bay ($p < 0.01$, Figs. 2b and S5d). Values of δ^{15}N-TDN did not differ between surface and bottom waters in Matsushima Bay ($p = 0.94$; Table S2 and Fig. S5e and f).

The spatial distribution pattern of δ^{15}N-TDN paralleled that of concentration (Fig. 5a). However, unlike the distribution of dissolved N species, δ^{15}N-TDN values at the SBP region were significantly higher than those in Sendai Bay ($p < 0.01$, Fig. 4d).

To further investigate how N species and their isotopic signatures vary along the salinity gradient, we analyzed the relationship between salinity and the concentrations

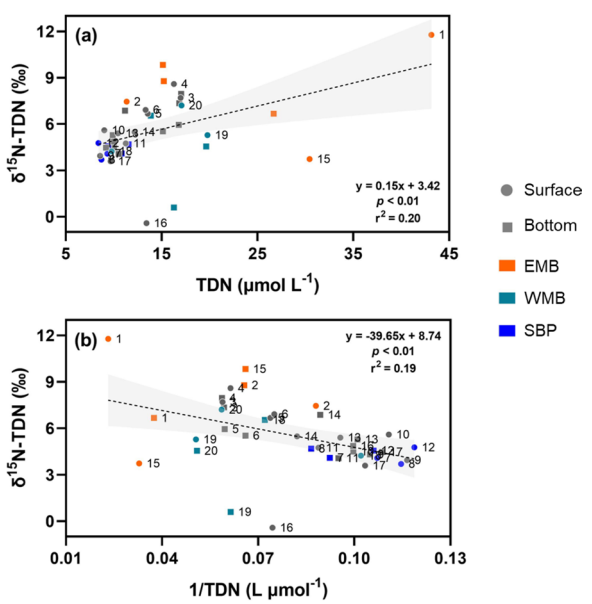

Fig. 5 Relation between TDN concentrations and δ^{15}N-TDN values in Matsushima Bay. **a** δ^{15}N-TDN values shown against TDN concentrations. **b** Keeling plot identifying the isotopic signatures of the source TDN running into Matsushima Bay. Keeling plot intercept: 8.74‰ within a 95% confidence interval of ± 2.25‰

and relative proportions of TDN components (NO_3^-, NH_4^+, and DON), as well as δ^{15}N-TDN (Fig. S10). As shown in Fig. S10, low-salinity samples were primarily associated with the EMB region due to high discharge (Tables S1), whereas elevated N concentrations and δ^{15}N-TDN values were also observed in regions such as WMB, where salinity remained relatively high due to limited discharge (Table S1). Consequently, while some trends can be seen along the salinity gradient, large variations in discharge and nutrient concentrations among rivers complicate the interpretation.

In addition, we compared salinity among the WMB, SBP, and Sendai Bay regions (Fig. S11) to examine how water mixing patterns correspond to spatial variations in N parameters. The results showed that, although salinity differed significantly between Matsushima Bay (including WMB and SBP) and Sendai Bay, there was no significant difference between WMB and SBP within Matsushima Bay ($p = 0.90$), despite large differences in nutrient concentrations and δ^{15}N-TDN values (Fig. 3).

These results suggest that salinity alone may not fully explain N distributions, whereas δ^{15}N-TDN still provides integrative information across regions.

3.3 DOC and $SUVA_{254}$

Findings indicate that DOC concentrations were higher in rivers (2.4–4.9 mg L^{-1}) than in seawater (Fig. 6a). In seawater, including Matsushima Bay (1.4 ± 0.2 mg L^{-1}) and Sendai Bay (1.7 ± 0.3 mg L^{-1}), DOC levels were distributed evenly (Fig. S9). Although DOC levels in Matsushima Bay were significantly lower than in Sendai Bay (0.4 ± 0.1 mg L^{-1}, $p < 0.01$), the absolute difference was small.

As a derivative parameter of DOC, $SUVA_{254}$ exhibited a different trend. $SUVA_{254}$ values were significantly higher in rivers (2.6–17.2 [L mg^{-1} C m^{-1}]) than in either Matsushima Bay (0.9–2.3 [L mg^{-1} C m^{-1}]) or Sendai Bay (0.4–1.1 [L mg^{-1} C m^{-1}) (Fig. 6b). The highest $SUVA_{254}$ values were found for the Sunaoshi-Teizan Canal (17.15 [L mg^{-1} C m^{-1}]); Fig. 6d). Therefore, the highest $SUVA_{254}$ values in Matsushima Bay were found in the surface water of MS1 (2.3 [L mg^{-1} C m^{-1}]), which is located at the WMB region of Matsushima Bay. The $SUVA_{254}$ values of Matsushima Bay were higher than those of Sendai Bay ($p < 0.01$), which contrasted with the trend observed for DOC concentrations (Fig. 6a and b).

4 Discussion

4.1 δ^{15}N-TDN as a promising parameter with high sensitivity for land-derived N

The distribution patterns of N components in Matsushima Bay and the surrounding regions provide valuable insights into the sources of N within this coastal

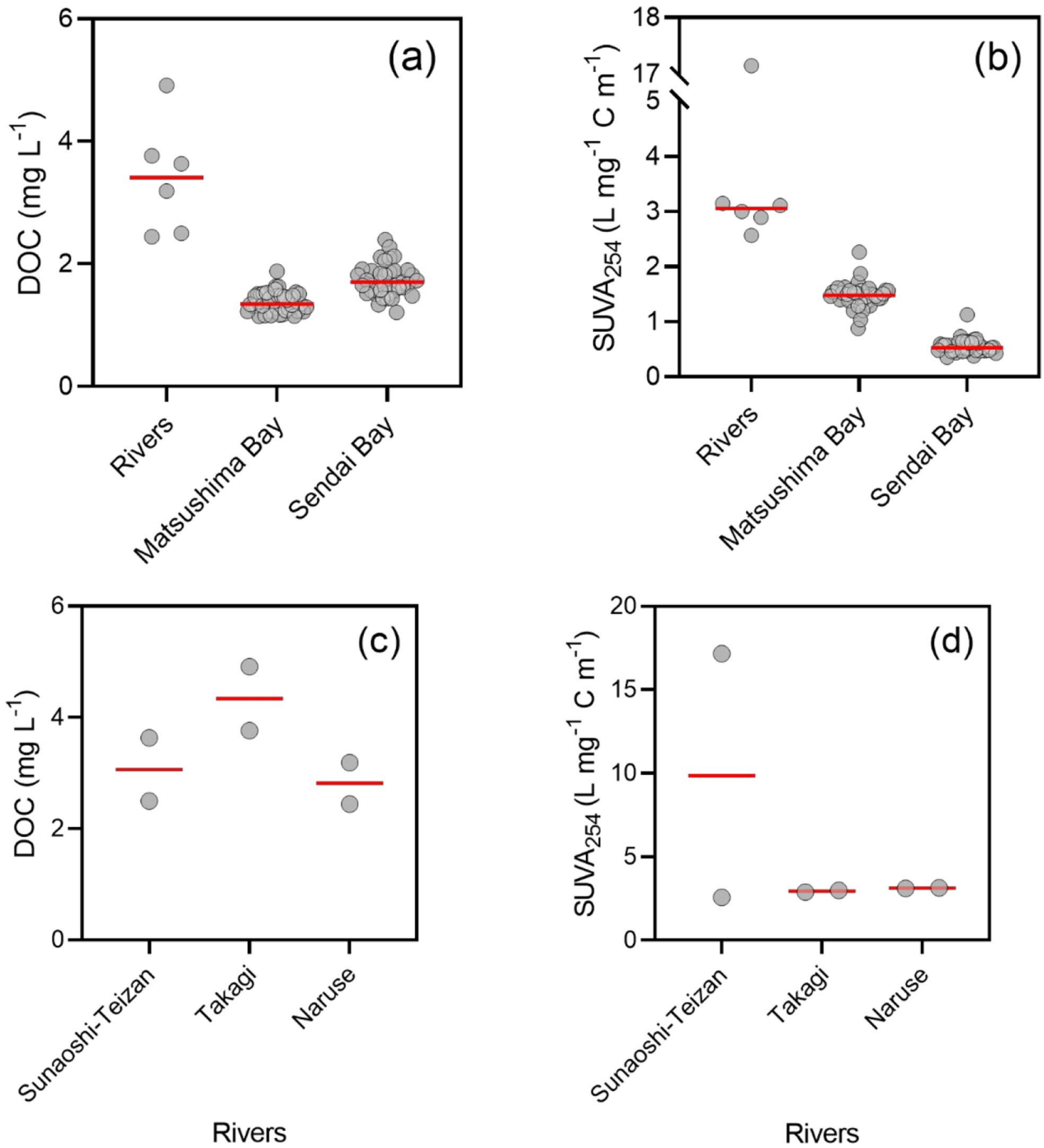

Fig. 6 Comparison of DOC concentrations and $SUVA_{254}$ values among rivers, Matsushima Bay, and Sendai Bay: **a** DOC concentrations across the rivers and bays; **b** $SUVA_{254}$ values across the rivers and bays; **c** DOC concentrations specifically for the Sunaoshi-Teizan Canal, Takagi, and Naruse rivers; and **d** $SUVA_{254}$ specifically for the Sunaoshi-Teizan Canal, Takagi, and Naruse rivers. Red lines represent the means of the DOC concentrations in panels **a** and **c**, and the medians of the $SUVA_{254}$ values in panels **b** and **d**

ecosystem. Our findings highlight the important roles of the Sunaoshi-Teizan Canal and Naruse River in contributing to N loading in Matsushima Bay (Figs. 3 and S4). Elevated concentrations of N at the EMB and WMB regions demonstrate their importance as nutrient sources.

However, analysis of nutrient concentrations offers only limited insights. Spatial analysis of TDN concentrations

reveals some limitations in using bulk nutrient data to trace riverine influences over longer distances (Fig. 4a–c). Despite high concentrations of river TDN, the surface water TDN concentrations in the WMB region (Fig. 1) decreased markedly (43.1, 30.4, and 11.4 μmol L^{-1}, respectively), probably because of low discharge rates of rivers (Table S1) and efficient seawater exchange. River water accounted for only 3.90% and 3.69%, respectively, of MS1 and MS15, based on salinity (Fig. S1c and d and Supplementary text). This rapid attenuation complicated the detection of riverine impacts in more distal parts of the bay. The minimal differences of concentrations in various N species between the SBP region of Matsushima Bay and Sendai Bay (Fig. 4a–c) imply that nutrient concentrations alone might not fully capture the spatial extent of riverine influence.

Unlike TDN concentrations, δ^{15}N-TDN values at the SBP region of Matsushima Bay remained markedly elevated compared to Sendai Bay, even in areas with a considerable degree of seawater mixing (Fig. 4d). Keeling plot analysis (Fig. 5b), which examines the relationship between the isotopic composition (δ^{15}N-TDN) and the inverse of TDN concentration to infer mixing between sources with distinct isotopic signatures (Keeling 1958), showed that the δ^{15}N-TDN variation in Matsushima Bay was shaped by the mixing between riverine N with high TDN concentration and δ^{15}N and seawater with low concentration and δ^{15}N (i.e., from Sendai Bay). The intercept of plot 8.7‰ (95% CI 6.5 to 11.0‰) for the Keeling plot, which was close to δ^{15}N of riverine TDN (6.7 to 14.7‰; Fig. 2b), strongly suggests riverine TDN running into Matsushima Bay. While our analysis primarily attributes the δ^{15}N-TDN variations to physical mixing, potential contributions from biological processes cannot be ruled out, such as isotopic enrichment of DIN through phytoplankton assimilation (Pennock et al. 1996) and the heterotrophic production of organic matter (Yamaguchi et al. 2017). These uncertainties highlight the need for further investigation.

Because of methodological difficulties, δ^{15}N-TDN data from aquatic ecosystems are quite scarce: Alkhatib et al. (2012) reported δ^{15}N-TDN (4.3–7.4‰) of bottom water in the St. Lawrence Estuary, whereas bottom water of eastern Bering shelf was 5–8‰ (Granger et al. 2011). Kharbush et al. (2023) reported δ^{15}N-TDN of eutrophic lake (Western Lake Erie) as 2.5–8.3‰. Apparently, δ^{15}N-TDN must be affected by δ^{15}N-DIN and δ^{15}N-DON, along with the relative contributions of DIN and DON to TDN. The dominant N in TDN was DON in Matsushima Bay (71 ± 18%; $n = 39$ without surface water at MS1 with quite high DIN concentration resulting in negative DON concentration estimated; Table S2). Because DON has a much lower turnover rate than DIN in aquatic ecosystems (Berman and Bronk 2003; Bronk et al. 2007) and DIN contributes only a small fraction to TDN in this study, the δ^{15}N-DON signature could still be preserved. Although DIN is rapidly produced and consumed, leading to variable δ^{15}N due to isotopic fractionation during turnover (Casciotti 2016), its limited contribution to TDN suggests that this variability may not strongly affect the δ^{15}N-DON signature. Although the δ^{15}N-DON signature cannot be interpreted either because of the high complexity of DON, we at least expect that the δ^{15}N-DON and then δ^{15}N-TDN of riverine N should be high compared with coastal N, as in the case for NO_3^- (Korth et al. 2014; Yu et al. 2015), because N discharged from agricultural field and wastewater plants is well-known to have high δ^{15}N (Costanzo et al. 2001; Fry 2006). Consequently, although it was not possible to interpret δ^{15}N-TDN explicitly, the clear difference in δ^{15}N-TDN between riverine N and coastal N demonstrates the utility of isotopic analysis in tracing riverine N and suggests that δ^{15}N-TDN can serve as a more sensitive and reliable indicator of riverine influence in coastal waters than the concentrations of various N species. In addition, it would be beneficial to incorporate other proxies (e.g., biogeochemical parameters) or estimates (such as N loading from each watershed) to support the conclusion that ^{15}N of TDN effectively indicates N and terrestrial organic matter inputs.

4.2 DOC and $SUVA_{254}$ illustrating the quality of dissolved organic matter (DOM)

Analysis of the DOC concentration and $SUVA_{254}$ further illustrates the complexity of organic matter dynamics in Matsushima Bay. Although DOC concentrations were higher in rivers (Fig. 6a), the minimal difference between Matsushima Bay and Sendai Bay indicated that riverine DOC input alone did not dominate the DOC pool in the bay. This observation might be attributed to the fact that river-derived DOC, which is 2.4–5.0 mg L^{-1} (Fig. 6c), was not much higher than in seawater, making its influence easily masked by other factors such as contributions from rivers directly flowing into Sendai Bay, i.e., the Abukuma, Natori, Yoshida, Naruse, and Kyu-Kitakami rivers (Kakehi et al. 2015) as well as rainfall and organic matter release and degradation by microorganisms. Therefore, as in the case of DIN and TDN concentrations, the DOC concentration data were ineffective to trace the riverine nutrient.

In contrast to DOC, $SUVA_{254}$ values provided clearer insights into the influence of riverine organic matter on the bay. High $SUVA_{254}$ indicate the high aromatic, terrestrial DOM (Weishaar et al. 2003), which is consistent with the data found from this study (Fig. 6b). The significantly higher $SUVA_{254}$ values in rivers and the elevated

values in Matsushima Bay compared to Sendai Bay suggest that the quality (Fig. 6b), rather than the quantity, of DOC is a better indicator of riverine influences. It is also noteworthy that $SUVA_{254}$, reflecting the aromaticity and potential terrestrial origin of DOC, remained elevated in Matsushima Bay, even where DOC concentrations were lower than in Sendai Bay. This highlights that, compared to DOC concentration, $SUVA_{254}$ serves as a more sensitive parameter for detecting riverine influences on coastal organic matter quality, as these riverine influences are often strongly affected by anthropogenic inputs such as organic matter from sewage treatment plants. In addition, this effectiveness of $SUVA_{254}$, possibly attributable to the slow alteration of DOM in this study, can be associated with the effectiveness of δ^{15}N-TDN because of the dominance of DON in TDN. Combined with these findings, DOM characteristics indirectly evaluated by δ^{15}N-TDN and $SUVA_{254}$ are demonstrably more effective for tracing the riverine nutrients when compared to DIN with high turnover, which might diminish information of the nutrients' origins.

5 Conclusion

This study emphasizes the superiority of qualitative indicators δ^{15}N-TDN and $SUVA_{254}$ for providing detailed insights into nutrient sources and distributions within Matsushima Bay. Analysis of δ^{15}N-TDN showed persistent differences in N isotopic signatures between the mouth of Matsushima Bay and Sendai Bay, despite their geographic proximity. In contrast, conventional concentration analyses were unable to capture the spatial extent and subtle nuances of riverine effects. Although δ^{15}N-TDN proved effective in this context, its effectiveness might be compromised in ecosystems with higher DIN concentrations, where δ^{15}N origin information can be diluted. Therefore, integrating qualitative indicators such as δ^{15}N-TDN and $SUVA_{254}$ with quantitative approaches offers a more comprehensive understanding of nutrient dynamics, especially in regions where the contribution from wastewater treatment plant effluent is significant, thereby improving the accuracy of nutrient status assessments and supporting more effective environmental management strategies for coastal ecosystems.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s40645-025-00787-3.

Additional file 1 (PDF 2271 KB)

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Author contributions
Conceptualization, developing methods, data interpretation, and writing of the manuscript (first draft) were done by M.J. and K.K. Sample collection was done by M.J., K.K., T.I., K.T., and K.O. Sample analysis was done by M.J., T.I., K.T., K.O., N.L.M., H.A., A.M., N.T., and K.K. Data processing and review: All authors contributed to data processing and reviewing of the manuscript.

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Data availability
The datasets supporting the conclusions of this article are included within the article and its Supplementary Materials.

Declarations

Competing interests
The authors declare that they have no competing interest.

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